

Content Page

List of Practical's and Progressive Assessment Sheet

Sr. No.	Title of the Practical	Page no.	Date of Performance	Date of Submission	Assessment Marks (25)	Dated sign. of Teacher	Remarks (If any)
1	Connect the Zener diode in the circuit and test its operation in forward and Reverse bias mode.						
2	Determine the voltage regulation by using Zener diode under variable input And output conditions.						
3	Make the input and output connections of UPS and measure the output voltage under online and offline mode.						
4	Make the input, output connections and check the operation of UPS under normal and overload condition.						
5	Test input /output characteristics of NPN transistor in CE configuration						
6	Check the switch ON and switch OFF condition of LED by using transistor.						
7	Measure temperature of Liquid using RTD (PT-100) transducer.						

8	Measure the resistance of LDR in varying light intensity.						
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Practical No. 1 : Connect the Zener diode in the circuit and test its operation in forward and reverse bias mode.

I Practical Significance:

Zener diodes are widely used as voltage references and as shunt regulators to regulate the voltage across small circuits. The student will be able to plot the forward and reverse characteristics of the Zener diode and measure the Zener voltage

II Industry/Employer Expected Outcome(s):

Electrical and Electronic industries use different types of DC power supplies with different voltage and current ratings. Zener diode works as a voltage regulator in DC power supply. The Employee should be able to test Zener diode for its ratings.

III Course Level Learning Outcome(s):

Use relevant diode in different Electronic circuits.

IV Laboratory Learning Outcome(s):

LLO Check the forward and reverse VI characteristics of Zener diode.

V Relevant Affective Domain related outcome(s):

1. Handle components and equipment carefully.
2. Follow the safety precautions.

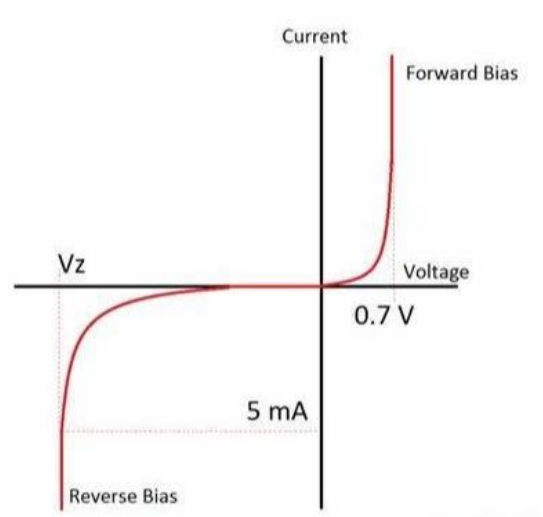
VI Relevant Theoretical Background (With diagrams if required):

Zener diode is formed by combining highly doped P and N semiconductor materials. It works on the principle of Zener breakdown and is normally operated in reverse breakdown region. In reverse breakdown region, high current flow through the diode leading to high power dissipation.

The Zener breakdown occurs when the electric field across the junction produced due to the reverse voltage is sufficiently high, this breaks covalent bonds. Thus a large numbers of carriers are generated which causes a more current to flow. This mechanism is called as Zener breakdown. After Zener breakdown the reverse current increases sharply but voltage across Zener diode remains constant. Zener resistance of a Zener diode is a ratio of reverse Zener voltage to the reverse Zener current.



Figure 1: Symbol of Zener diode



V-I Characteristics of Zener diode

II Actual Circuit diagram used in laboratory with equipment Specifications:

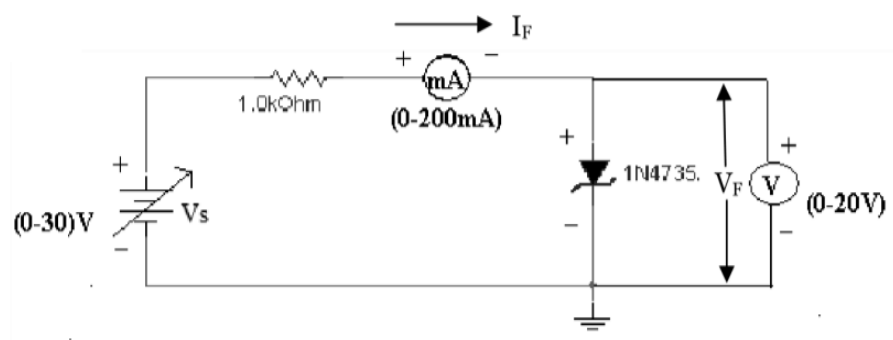


Figure 2: Zener diode in forward bias

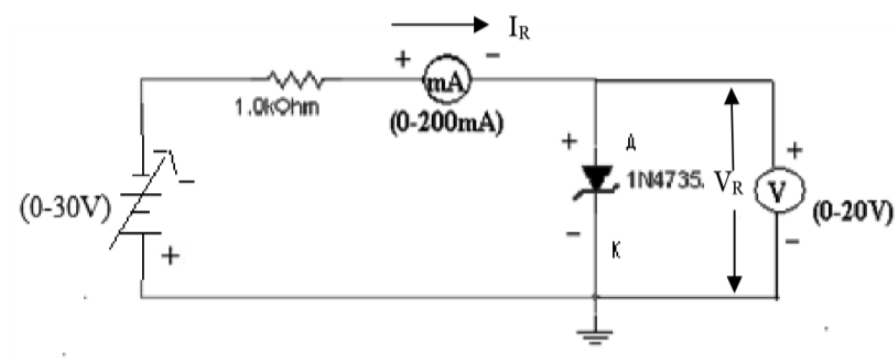


Figure 3: Zener diode in reverse bias

II Required Resources/apparatus/equipment with specification:

Sr.No.	Instrument /Components	Specification	Quantity

1	Digital Multimeter	Digital Multimeter	2
2	DC Regulated power supply	Variable DC power supply 0-30V,2A,SC protection, Display for voltage and current.	1
3	Voltmeter	0-20V	1
4	Ammeter	(0-200ma,0-200 μ A)	1
5	Breadboard	5.5CM X 17CM	1
6	Zener Diode	IN4735(or any other equivalent diode)	1
7	Resistor	1K Ω (0.5watts/0.25watta)	1
8	Connecting wires	Single strand Teflon coating(0.6mmdiameter)	As per requirement

II Precautions to be followed:

1. Do not switch ON the power supply unless you have checked the circuit connections as per the circuit diagram.
2. While doing the experiment do not exceed the input voltage of the Zener diode beyond the rated voltage of diode. This may lead to damaging of the diode.
3. Connect voltmeter and ammeter in correct polarities as shown in the circuit diagram.

III Procedure:

1. Connect the circuit as shown in figure2.
2. Switch ON the power supply.
3. Record the forward voltage and forward current in the observation table no1.
4. Increase the input voltage in step of 0.1 V
5. Record the forward voltage and forward current in the observation table no1.
6. Repeat steps 4to 5till 1V is reached.
7. Plot the graph for the forward bias characteristics of Zener diode by taking forward voltage on X- axis and forward current on Y-axis.
8. Connect the circuit as shown in figure3.
9. Vary input voltage gradually in steps of 1Vup to12V.
10. Record the corresponding readings of reverse voltage and reverse current in the observation table no2.
11. Plot the graph for the reversebias characteristics of Zener diode by taking V_{RON} X- axis and I_{RON} Y-axis.

VII Required Resources/apparatus/equipment with specifications:

S. No	Instruments/Components	Suggested broad specification	Quantity
1			

2			
3			

VIII Actual procedure followed:

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XIII Observations and Calculations:

Table1: Measurement of V_F and I_F

SR.NO	V_F (volt)	I_F (mA)
1		
2		
3		
4		
5		
6		
7		
8		
9		

Table2: Measurement of V_R and I_R

SR.NO	$V_R(\text{volt})$	$I_R(\text{mA})$
1		
2		
3		
4		
5		
6		
7		
8		
9		

Calculations: $R_F = \Delta V_F / \Delta I_F \text{ ohm}$

$$R_Z = \Delta V_R / \Delta I_R \text{ ohm}$$

XIV Results:

1. Zener break down voltage =V.
2. Zener resistance = Ω

XV Interpretation of Results:

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XVI Conclusions & Recommendations:

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XVII Practical Related Questions:

1. What is the value of Zener voltage for given zener diode?
2. What is the maximum value of reverse current for given zener diode?
3. What is the effect on voltage across zener diode and current flowing through it, when reverse voltage across it is more than breakdown voltage?
4. What portion of zener diode characteristics is most useful for voltage regulation applications?

[Space for Answers]

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Marks Obtained			Dated signature of teacher
Process Related (15)	Product Related(10)	Total (25)	

X-axis -

Y-axis -

Practical No.2 :Determine the voltage regulation by using zener diode under variable input and output conditions.

I Practical Significance:

It is required to provide regulated power supply to various circuits and Integrated Circuits. Zener diodes have a primary application as a voltage regulator. Various electronic equipment's and circuits require regulated power supply which can be provided by Zener voltage regulator. Student should be able to test zener as a voltage regulator for variable input and variable load.

II Industry/Employer Expected Outcome(s):

Electrical and Electronic industries use different types of DC power supplies with different voltage and current ratings. Zener diode works as a voltage regulator in DC power supply. Depending on the input voltage and required output voltage zener diode should be selected.

III Course Level Learning Outcome(s):

Use relevant diode in different Electronic circuits.

IV Laboratory Learning Outcome(s):

Find the voltage regulation of zener diode.

V Relevant Affective Domain related outcome(s):

1. Carefully handling of components and circuits.
2. Visually aesthetic connections.

VI Relevant Theoretical Background(With diagrams if required):

Zener diode is designed to operate in the breakdown region. It is possible to construct Zener diode with required breakdown voltage in reverse bias condition. After breakdown, Zener diode acts as a constant voltage source i.e. if the applied reverse voltage exceeds the Zener voltage; it keeps the voltage across the device constant. Since it acts as a constant voltage regulator i.e. it keeps the output voltage constant irrespective of changes in load current or changes in input voltage.

VII Actual Circuit diagram used in laboratory with equipment Specifications:

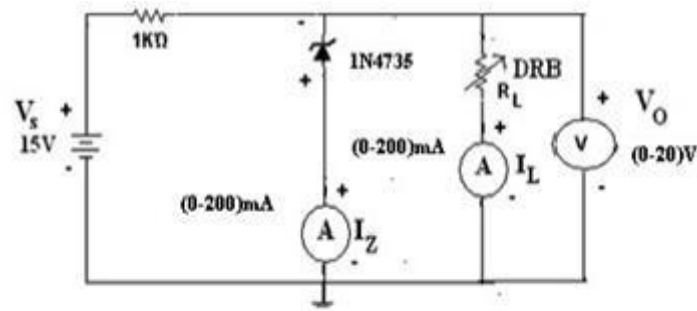


Figure 1: Load Regulation

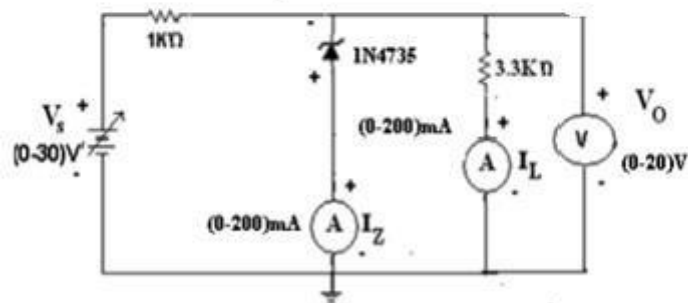


Figure 2: Line Regulation

VIII Required Resources/apparatus/equipment with specification:

S.No.	Instrument /Components	Specification	Quantity
1	Digital Multimeter	3½digit display	1
2	DC Regulated power supply	Variable DC power supply 0-30V,2A,display for Voltage and current.	1
3	ZenerDiode	1N4735(or any other equivalent diode)	1
4	DC Ammeter	0-200mA	02
5	Potentiometer	10KΩ	01
6	Resistor	1KΩ(0.5watts/0.25watts)	1
7	Connecting wires	Single strand Teflon coating(0.6mm diameter)	As per requirement

IX Precautions to be followed:

1. Do not switch ON the power supply unless the circuit connection are checked as per the circuit diagram.
2. See the data sheet to know the reverse break down voltage of the given diode before Starting the experiment.
3. Connect Voltmeters/Ammeters in correct polarities as shown in the circuit diagram
4. Switch OFF the power supply after taking readings.

X Procedure:

1. Connect the circuit as per the circuit diagram.
2. Switch ON the power supply.

Load Regulation(Keep the Input voltage at 5V-10V)

1. By changing the Load Resistance R_L , measure the corresponding output(Voltmeter)voltage.
2. Measure the current in the two ammeters to measure zener current I_z and Load current I_L

Line Regulation

Keep the Load resistance R_L constant. Vary the input supply V_s & note down the corresponding output voltage.

Graph

Plot the graph of Load current I_L (X-axis) Versus Load voltage V_o (Y-axis). Plot a graph between Input voltage V_s (X-axis) Versus Output Voltage V_o (Y-axis)

XI Required Resources/apparatus/equipment with specifications:

S. No	Instruments/Components	Suggested broad specification	Quantity
1			
2			
3			
4			

XII Actual procedure followed:

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XIII Observations and Calculations:

a. Load Regulation
Input Voltage $V_S =$ -----

Table: 1

Sr. No	$I_Z(\text{mA})$	$I_L(\text{mA})$	$V_O(\text{V})$

Calculation: No Load Voltage V_{NL} - Output voltage across R_L when Load current is minimum. Full load Voltage V_{FL} - Output voltage across R_L when Load current is maximum.

b. Line Regulation

le: 2

Sr. No	V _s (V)	I _Z (mA)	I _L (mA)	V _O (V)

Calculations:

i. Load Regulation = $\frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$ (From Table: 1)

ii. Line Regulation = $\frac{\Delta V_O}{\Delta V_S} \times 100$ (From Table: 2)

XIV Results:

1. Zener breakdown voltage=.....
2. Forward resistance of zener diode=.....Ω

XV Interpretation of Results:

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XVI Conclusions &Recommendations:

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XVII Practical Related Questions:

1. Give the value of input voltage when Zener current starts increasing?
2. For what value of Load resistance the Load current is Minimum?
3. For what value of Load resistance the Load current is Maximum?
4. Define zener breakdown?

[Space for Answers]

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XVIII References/suggestions for further reading;includes websites:

<https://www.youtube.com/watch?v=itzPT3UbC1I>

S.No.	Title of Book/Website	Author	Publication
1	Applied Electronics	R.S.Sedha	S.ChandandCo.,NewDelhi2008,ISBN978-8121927833
2	Principles of Electronics	V.K.Mehta	S.Chand and Co.,Ram Nagar,New Delhi-110055,11 th Edition,2014,ISBN978-812-192405

Marks Obtained			Dated signature of teacher
Process Related (15)	Product Related(10)	Total (25)	

5043

X-axis -

Y-axis -

Practical No.3 : Make the input and output connections of UPS and measure the output voltage under online and offline mode

I Practical Significance:

Uninterruptible Power Supplies (UPS) have important role in various applications where a reliable and continuous power supply is required.

II Industry/Employer Expected Outcome(s):

In industrial, domestic, commercial applications, continuous AC mains supply is needed. The employee is expected to select UPS of suitable ratings and test it under online and offline mode.

III Course Level Learning Outcome(s):

Use relevant diode in different Electronic circuits.

IV Laboratory Learning Outcome(s):

Check the operation of UPS under online and offline mode.

V Relevant Affective Domain related outcome(s):

1. Handle the equipments and components carefully.
2. Follow safety precautions.

VI Relevant Theoretical Background (With diagrams if required):

The purpose of a UPS is to provide emergency power (usually by a lead/acid battery) to a load when it senses that the input power source has failed. They are different from emergency power systems or standby generators because they provide near-instantaneous protection from power interruption by using a battery (which can be a super capacitor or flywheel). The battery itself usually has a short runtime (about 5-20 minutes), but it should be enough to either save all that precious data/progress that you have made, gracefully shut everything down, or fix the problem that caused the outage. A UPS can be used to protect hardware like data centers, computers, and other electrical equipment where an unexpected surge/sag can cause serious problems like data loss, business disruption, and even injuries or fatalities.

Types of Uninterruptible Power Supplies

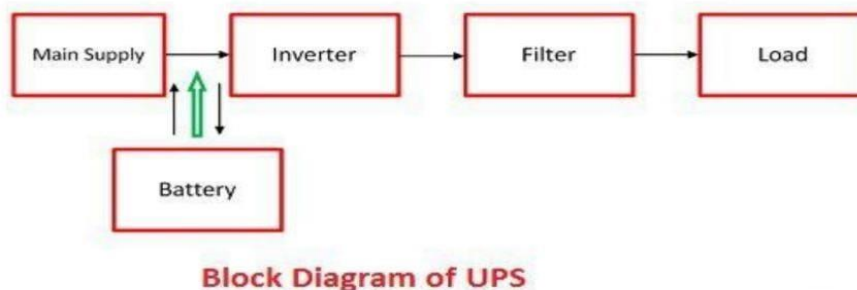
There are three types of uninterruptible power supplies: static, dynamic (rotary), and hybrid. Static uses power electronic converters, dynamic uses electromagnetic engines (generators and motor), and hybrid uses a combination of both static and dynamic.

1. Offline/Stand by Uninterruptible Power Supply

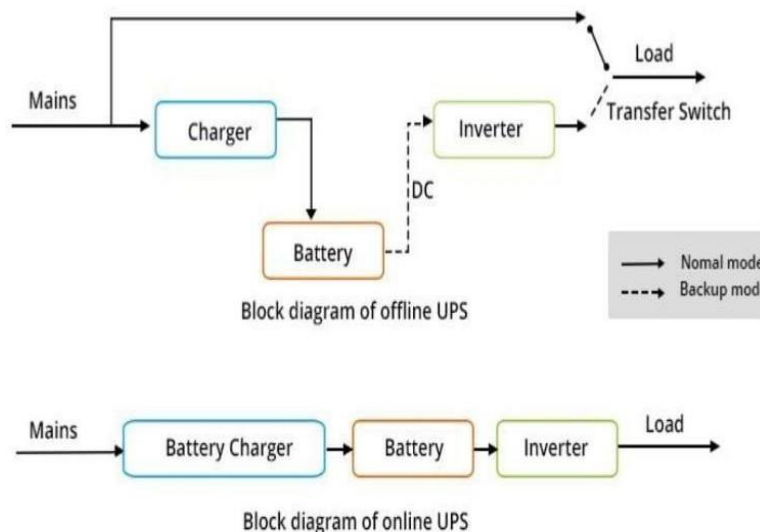
The offline/standby UPS is the most basic out of the three. It provides light surge protection and battery back-up. During normal operations, it gets its power from its main power source (generally an AC outlet). Once it senses that the main power source goes beyond acceptable limits or fails, it switches to the "offline/standby" battery where it will then go to the DC/AC inverter power source and battery as such, there will be a small transfer time between the main

2. Online/Double Conversion Uninterruptible Power Supply

The online/double conversion UPS differs from the offline/standby as the DC/AC inverter is always connected on. This means there will be no transfer time between the main power source and battery, providing greater protection against spikes, sags, electrical noise, and complete power failure.



VII Actual Circuit diagram used in laboratory with equipment Specifications:



VIII Required Resources/apparatus/equipment with specification:

Sr.No.	Name of Resource	Suggested Specifications	Quantity
1.	UPS	Input voltage-230VAC $\pm$ 15%single phase Output Voltage- 230V $\pm$ 1% single phase Power capacity- 1kVA Frequency-50 Hz $\pm$ 0.5Hz	1
2.	Digital Multi meter	DMM:3½ digit display	1

VIX Precautions to be followed:

1. Do not install UPS in an out door area that is open to direct sunlight, moisture, thunderstorms or other drastic climatic conditions.
2. Only a skilled technician should do battery replacement.
3. Do not open the UPS over unnecessarily.
4. If at all, you have to open the cover for checking any fault, it should be done only after switching the UPS off and disconnecting it from the mains.

X Procedure:**Mains ON:**

1. Measure AC input voltage with the help of multi meter.
2. Measure AC output voltage with the help of multi meter.
3. Measure Battery Full Charge DC Voltage with the help of multi meter.

Mains OFF:

1. Measure AC input Voltage with the help of multi meter.
2. Measure AC output Voltage with the help of multi meter.
3. Measure Battery DC voltage with the help of multi meter.

XI Required Resources/apparatus/equipment with specifications:

S. No	Instruments/Components	Suggested broad specification	Quantity
1			
2			
3			
4			

XII Actual procedure followed:

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XIII Observations and Calculations:

Sr.No.	Operating conditions	Input Voltage	Output Voltage
1.	Online Mode		
2.	Offline Mode		

XIV Results:

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XV Interpretation of Results:

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XVI Conclusions & Recommendations:

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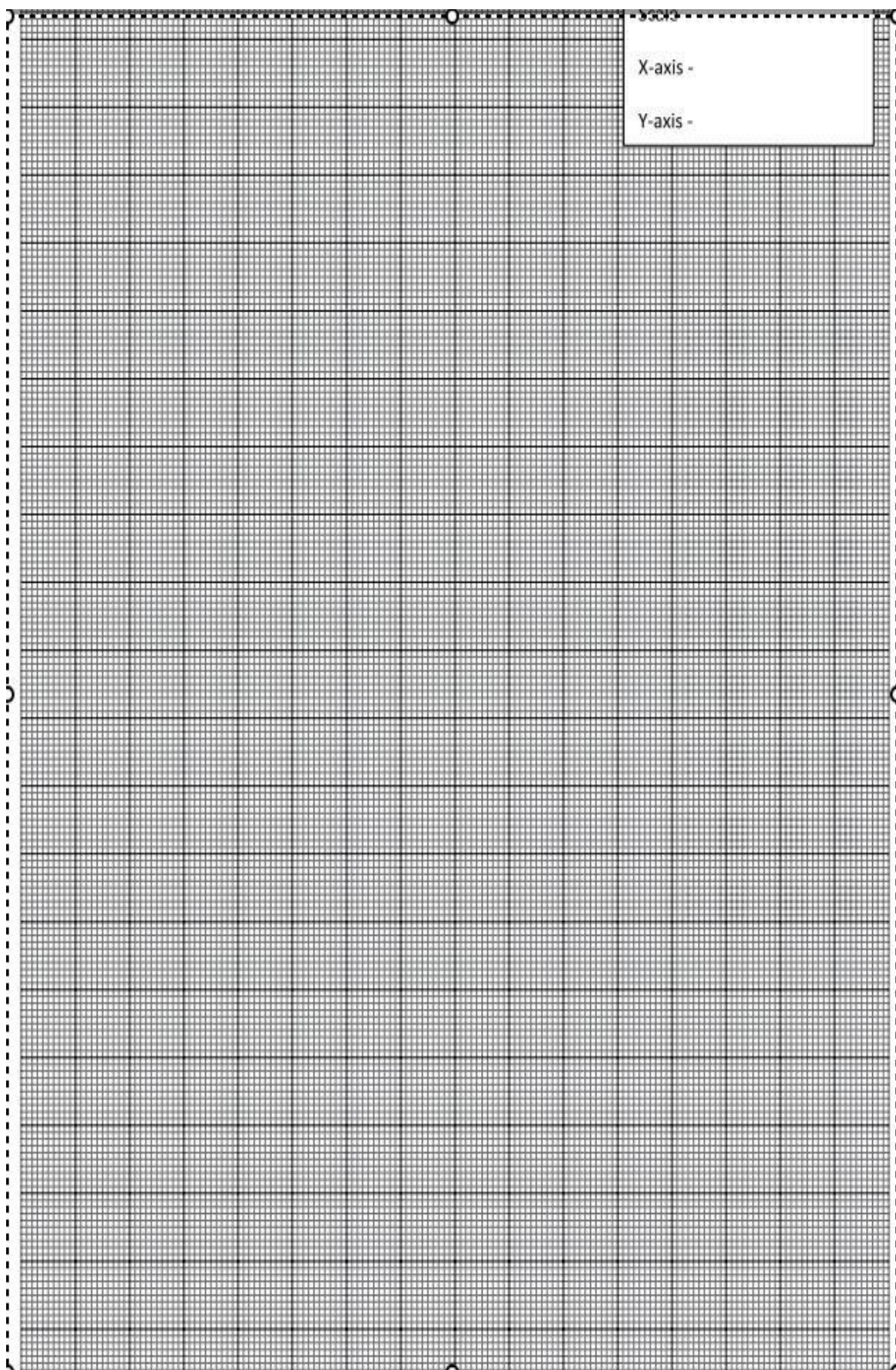
XVII Practical Related Questions:

1. What is mean by UPS?
2. List types of UPS.
3. List applications of UPS.
4. State the need of UPS.

[Space for Answers]

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Marks Obtained			Dated signature of teacher
Process Related (15)	Product Related(10)	Total (25)	



X-axis -

Y-axis -

Practical No.4 : Make the input and output connections and check the operation of UPS under normal and overload condition

I Practical Significance:

Uninterruptible Power Supplies (UPS) have vital role in various applications where a reliable and continuous power supply is essential.

II Industry/Employer Expected Outcome(s):

In industrial, domestic and commercial applications continuous AC mains supply is needed. This can be provided by UPS. The employee is expected to select UPS of suitable ratings and test it under normal and overload conditions.

III Course Level Learning Outcome(s):

Use relevant diode in different Electronic circuits.

IV Laboratory Learning Outcome(s):

Check the abnormal and normal operation of UPS.

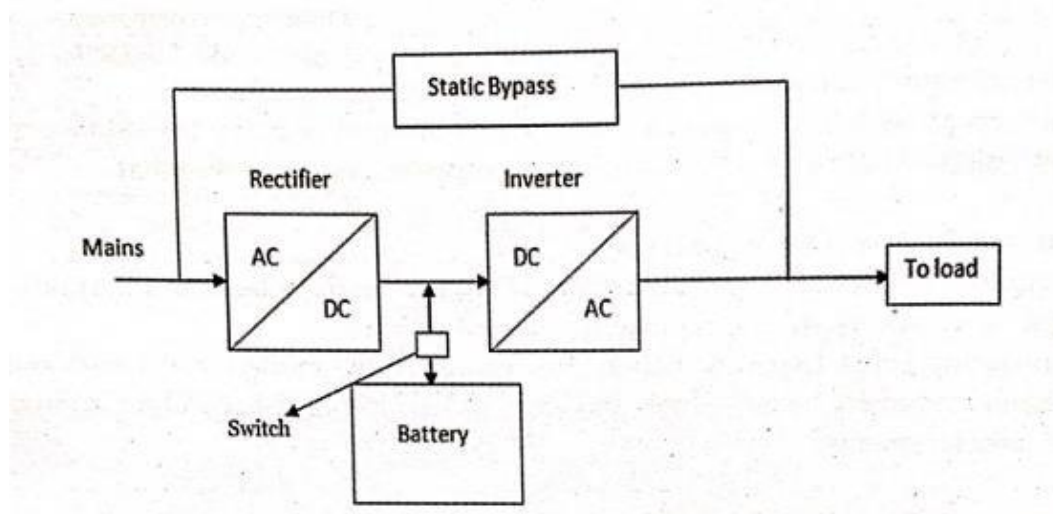
V Relevant Affective Domain related outcome(s):

1. Handle the equipment's and components carefully.
2. Follow safety precautions.

VI Relevant Theoretical Background (With diagrams if required):

In normal operation, a double-conversion UPS continually processes power twice. If the AC input supply falls out of predefined limits, however, the input rectifier shuts off and the output inverter begins drawing power from the battery instead. If you overload your UPS, it will either shut down or the load will be transferred to by-pass until the condition is reversed. When this happens, all of the devices connected to it will lose power.

VII Actual Circuit diagram used in laboratory with equipment Specifications:



VIII Required Resources/apparatus/equipment with specification:

Sr. No.	Name of Resource	Suggested Specifications	Quantity
1.	UPS	Input voltage-230VAC $\pm$ 15% single phase Output Voltage- 230V $\pm$ 1% single phase Power capacity- 1kVA Frequency-50Hz $\pm$ 0.5Hz	1
2.	Digital Multi meter	DMM:3½ digit display	1

IX Precautions to be followed:

1. Do not install UPS in an out door area that is open to direct sunlight, moisture, thunder storms or other drastic climatic conditions.
2. Only a skilled technician should do battery replacement.
3. Do not open the UPS cover unnecessarily.
4. If at all, you have to open the cover for checking any fault, it should be done only after switching the UPS off and disconnecting it from the mains.

X Procedure:

a. When the load is OFF

1. Measure output voltage of UPS with the help of multi meter.
2. Measure Backup time with the help of stop watch.

b. When the load is overload

1. Measure output voltage of UPS with the help of multi meter.
2. Measure Backup time with the help of stop watch.

XI Required Resources/apparatus/equipment with specifications:

S. No	Instruments/Components	Suggested broad specification	Quantity
1			
2			
3			
4			

XII Actual procedure followed:

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XIII Observations and Calculations:

When the load is OFF

1. Output voltage of UPS Volts
2. Back up time=min.

When the load is overload:

1. Output voltage of UPS Volts
2. Backup time.....min.

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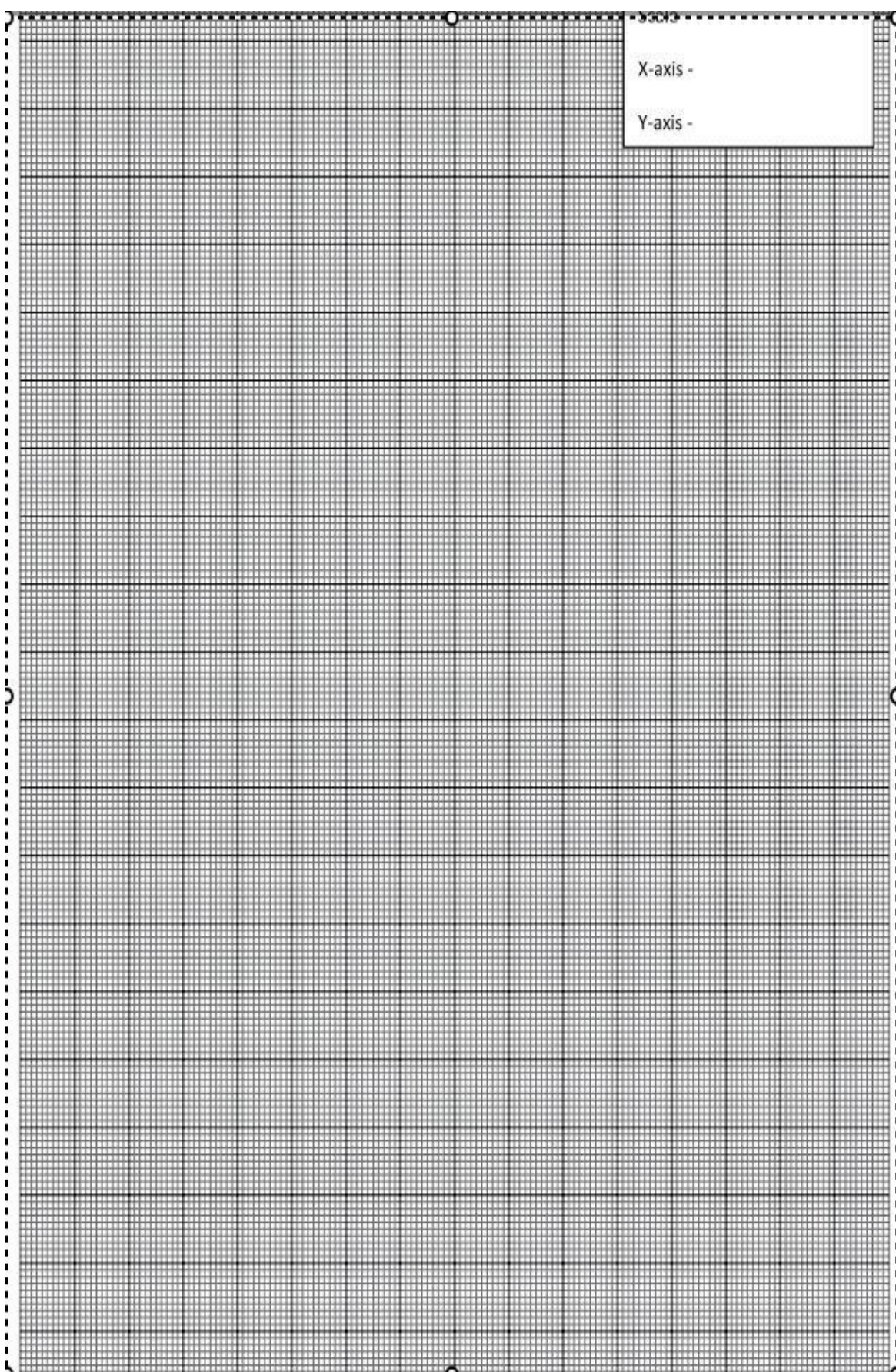
1. Write the technical specifications of the given UPS.
2. Write the type of battery used in UPS.
3. State the backup time of the UPS used.
4. List applications of UPS.

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XVIII References/suggestions for further reading; includes websites:

1. UPS Working Principle and Types- Offline and Online UPS Systems
2. MicrosoftWord-MpBe002c_EnergieeffizienzvonUSV-Anlagen_Schlussbericht.doc

Marks Obtained			Dated signature of teacher
Process Related (15)	Product Related(10)	Total (25)	



X-axis -

Y-axis -

PracticalNo.5 : Test input/output characteristics of NPN transistor in CEC on figuration.

I Practical Significance:

A BJT is commonly used as an amplifier. Common Emitter (CE) mode is the universal mode of operation for a BJT. All types of amplifications can be performed using CE mode with suitable biasing. Common-emitter amplifiers are also used in radio frequency circuits..

II Industry/Employer Expected Outcome(s):

Amplifier circuits are used in all electronic equipment used in industries. Employee should be able select suitable type of transistor and configuration for given application.

III Course Level Learning Outcome(s):

Use BJT and FET in various Electronic circuits.

IV Laboratory Learning Outcome(s):

Check the operation of NPN Transistor under CE Configuration.

V Relevant Affective Domain related outcome(s):

1. Handle components and equipment carefully.
2. Work in team.

VI Relevant Theoretical Background(With diagrams if required):

BJT is called as Bipolar Junction transistor. It has 3 terminals namely emitter, base and collector. It is called bipolar device because current through it is due to free electrons and holes. A transistor can be in any of the three configurations namely common base, common emitter and common collector.

The relation between of α , β and γ of CB,CE&CC are

$$\alpha = \beta / 1 + \beta$$

$$\beta = \alpha / 1 - \alpha$$

$$\gamma = 1 + \beta = 1 / 1 - \alpha$$

In CE configuration base will be the input node and collector will be output node. Emitter is common to both input and output and hence the name common emitter configuration. A transistor in CE configuration is used widely as an amplifier

Symbol:

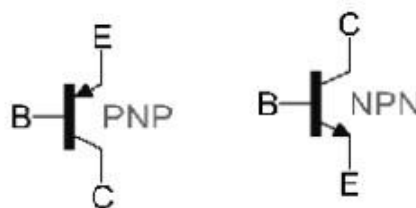


Figure 1: Symbol of BJT

Input Characteristics:

This curve gives the relationship between input current (I_B) and input voltage (V_{BE}) for constant output voltage (V_{CE}). By varying V_{BE} for constant V_{CE} it may be noted that below knee voltage current is very small. Beyond knee voltage, the base current (I_B) increases with increase in V_{BE} for constant V_{CE} .

Input characteristics may be used to determine the value of common emitter transistor a.c. input resistance r_i . It is the ratio of change in base to emitter voltage (ΔV_{BE}) to resulting change in base current (I_B) at a constant collector to emitter voltage (V_{CE}).

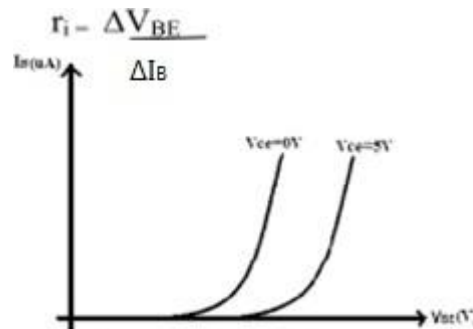


Figure 2: Input Characteristics of BJT in CE mode

Output Characteristic:

This curve gives the relationship between output current (I_C) and output voltage (V_{CE}) for constant base current (I_B).

The output characteristics are divided into three regions:

Cut off region: Transistor act as off switch.

Saturation region: Transistor act as on switch.

Active region: Transistor act as amplifier.

Output Characteristic may be used to determine the value of common emitter transistor a.c. output resistance r_o it is the ratio of change in collector to emitter voltage (ΔV_{CE}) to resulting change in collector current (ΔI_C) at a constant base current (I_B).

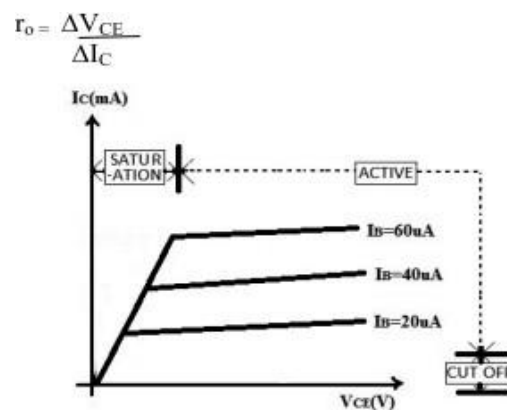


Figure 3: Output Characteristics of BJT in CE mode

VII Actual Circuit diagram used in laboratory with equipment Specification:

(a) Sample

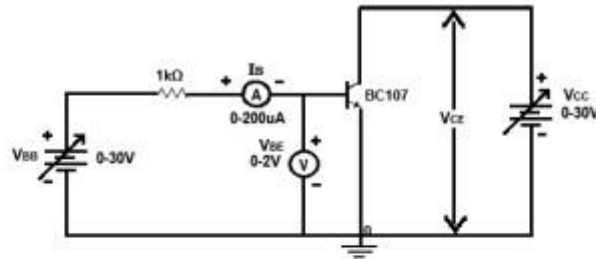


Figure: 3 Circuit diagram for Input Characteristics

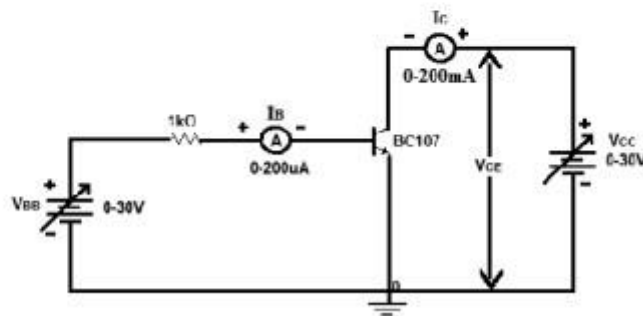


Figure 4: Circuit diagram for Output Characteristics

VIII Required Resources/apparatus/equipment with specification:

Sr.No.	Name of Resource	Suggested Specification	Quantity
1	DC power supply	0-30V	2 no.
2	DC Voltmeter	(0-2V),(0-20V)	1 no.
3	DC Ammeter	(0-50milliamps),(0-500μA)	1 no.
4	Transistor	BC107	1 no.
5	Resistor	1KΩ	1 no.
6	Any other		

IX Precautions to be followed:

1. Care should be taken while handling terminals of components.
2. Select proper range and mode of ammeter and voltmeter.
3. Connect wires tightly while building circuit.
4. Show the connections to concerned teacher and then switch ON the power supply.

X Procedure:

Part I Input characteristics:

1. Connect the electrical circuit as shown in Figure 3.
2. Select suitable range of milli-ammeter, voltmeter and power supply.
3. Switch on the power supply and adjust the voltage $V_{CE}=2V$ by varying V_{CC}
4. Vary the input voltage V_{BE} in steps of $0.1V$ and measure the current I_B for each Step.
5. Repeat the steps 3 and 4 for V_{CE} values of $5V$ and $10V$.
6. Tabulate the readings

Part II Output characteristics:

1. Connect ammeter and adjust base current I_B .
2. Select suitable range of milli-ammeter, voltmeter and power supply.
3. Switch on the power supply and apply a constant current $I_B=20 \mu A$.
4. Vary V_{CE} from 0 to $10V$ in steps of 0.5 volts. Measure corresponding collector current I_C for each step.
5. Repeat steps 9 and 10 for various values of $I_B=30 \mu A$ and $40 \mu A$
6. Tabulate the readings

Graph

Plot a graph of $V_{BE}(V)$ (X-axis) versus $I_B(\mu A)$ (Y-axis) for different V_{CE} voltages.

Plot a graph of $V_{CE}(V)$ (X-axis) versus $I_C(mA)$ (Y-axis) for different $I_B(\mu A)$ currents.

Calculate dynamic input resistance using the formula given below

$$r_i = \left. \frac{\Delta V_{BE}}{\Delta I_B} \right|_{V_{CE}} = \text{-----}$$

Calculate dynamic output resistance using the formula given below.

$$r_o = \left. \frac{\Delta V_{CE}}{\Delta I_C} \right|_{I_C} = \text{-----}$$

XI Required Resources/apparatus/equipment with specifications:

S. No	Instruments/Components	Suggested broad specification	Quantity
1			
2			
3			
4			

XII Actual procedure followed:

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XIII Observations and Calculations:

Table 1: Input characteristics

S.No.	$V_{CE} = \dots\dots V$		$V_{CE} = \dots\dots V$		$V_{CE} = \dots\dots V$	
	$V_{BE} (V)$	$I_B (\mu A)$	$V_{BE} (V)$	$I_B (\mu A)$	$V_{BE} (V)$	$I_B (\mu A)$
1						
2						
3						
4						
5						
6						

Table 2: Output characteristics

S.No.	$I_B = \dots\dots\dots \mu A$		$I_B = \dots\dots\dots \mu A$		$I_B = \dots\dots\dots \mu A$	
	$V_{CE} (V)$	$I_C (mA)$	$V_{CE} (V)$	$I_C (mA)$	$V_{CE} (V)$	$I_C (mA)$
1						
2						
3						
4						
5						
6						

Calculations:

$$r_i = \frac{\Delta V_{BE}}{\Delta I_B}$$
$$r_o = \frac{\Delta V_{CE}}{\Delta I_C}$$
$$\beta = \frac{I_C}{I_B}$$

XIV Results:

- 1. Input resistance= Ω
- 2. Output resistance=..... Ω
- 3. Current amplification factor β =.....

XV Interpretation of Results:

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XVI Conclusions & Recommendations:

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XVII Practical Related Questions:

1. Write the steps to identify emitter, base and collector terminals of given transistor.
2. State the range of ammeter and voltmeter selected
3. The BJT has $I_B=10\text{ }\mu\text{A}$, $I_{CO}=1\text{ }\mu\text{A}$ what is collector current I_C ? (Take value of β obtained from the graph)?

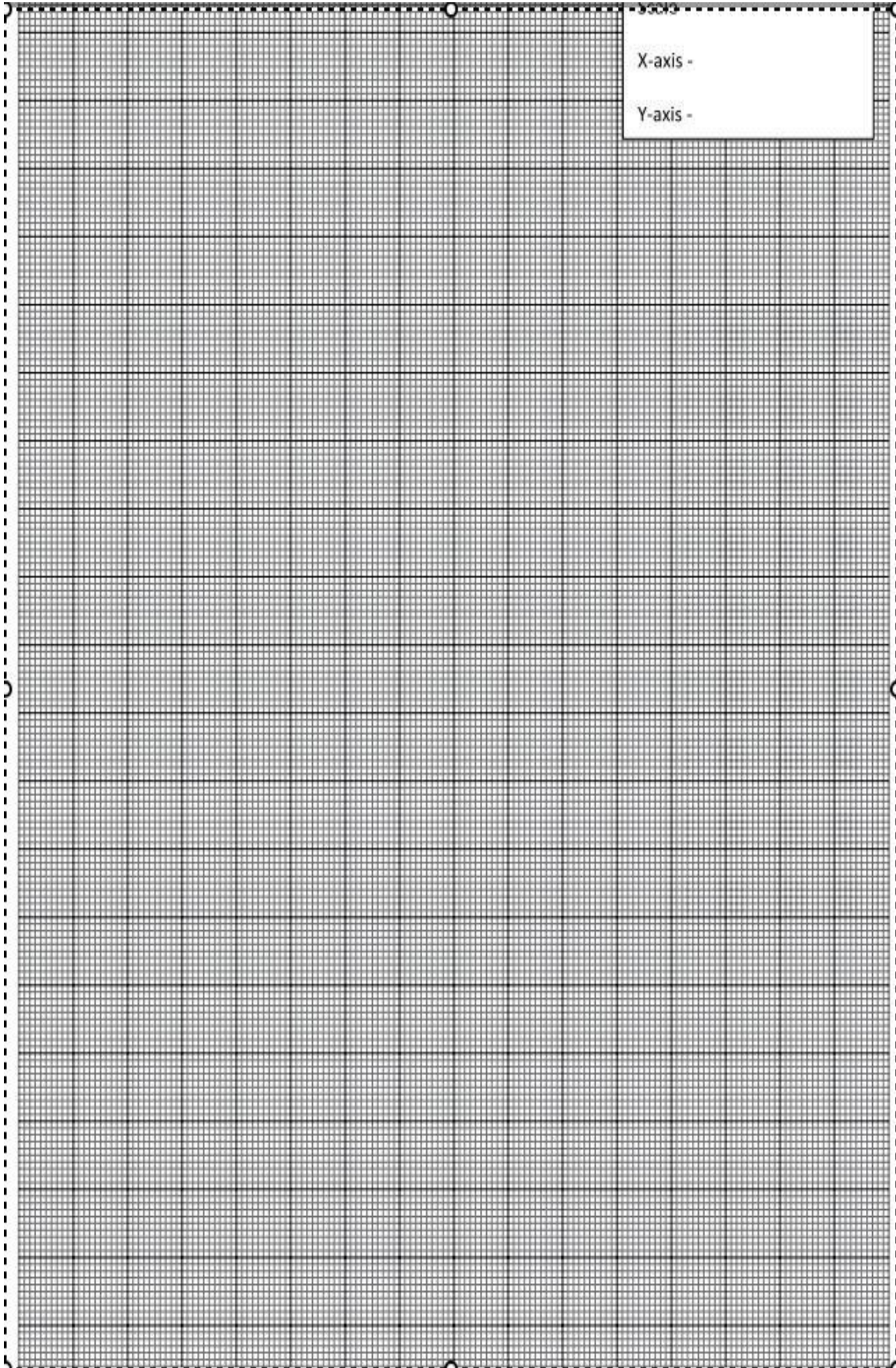
[Space for Answers]

This image shows a full page of primary-ruled paper. It features multiple sets of three horizontal dotted lines, which are commonly used in elementary school writing to guide letter height and placement. The lines are evenly spaced across the entire page, providing a template for handwriting practice. There is no text or other markings on the paper.

XVIII References/ suggestions for further reading; includes websites:

S. No.	Title of Book / Website	Author	Publication
1	Applied Electronics	R.S.Sedha	S. Chand and Co., New Delhi 2008,ISBN 978-8121927833
2	Principles of Electronics	V.K.Mehta.	S. Chand and Co., Ram Nagar ,New Delhi-110055,11 th Edition,2014.ISBN 978812192405
3	Web References		a. https://www.electrical4u.com/transistor-characteristics/ b. http://nptel.ac.in/courses/117107095/10

Marks Obtained			Dated signature of teacher
Process Related (15)	Product Related(10)	Total (25)	



Practical No.6 :Check the switch ON and switch OFF condition of LED by using transistor.

I Practical Significance:

The purpose of the Practical is to design and analyse the operation of transistor as a switch. Also, to test a suitable driver circuit and multi vibrators circuits for a given application using transistor.

II Industry/Employer Expected Outcome(s):

Many industrial application uses high frequency switching technique in electronic instruments. Employee should select and test different types of transistors as a high frequency switch.

III Course Level Learning Outcome(s):

Use BJT and FET in various electronic circuits.

IV Laboratory Learning Outcome(s):

Check operation of transistor for ON and OFF condition.

V Relevant Affective Domain related outcome(s):

1. Handle components and equipment carefully.
2. Work in team

VI Relevant Theoretical Background (With diagrams if required):

A transistor can function as a single-pole single-throw switch controlled by an electronic signal driving the base terminal. When the control signal on the base terminal turns the transistor off, it acts like an open switch. When the control signal on the base terminal turns the transistor on, it acts like a closed switch. When transistor is used for switching, it is in one of two states on or off . In the off state, the base bias current is zero and the transistor is in cut off region. In the on state, the base bias current is set large enough to drive the transistor into saturation region.

VII Actual Circuit diagram used in laboratory with equipment Specifications:

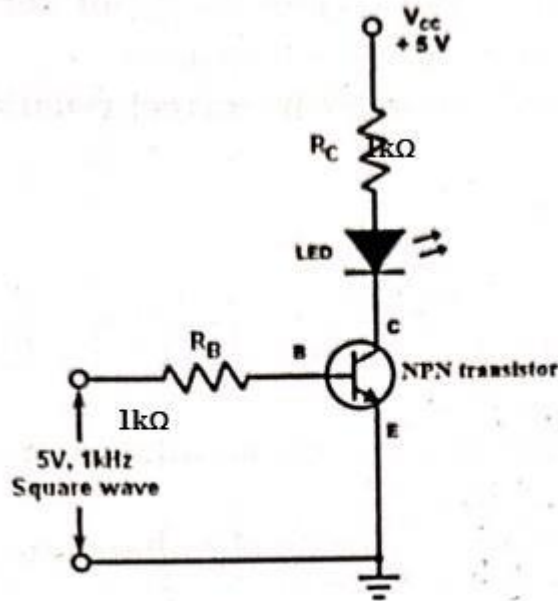


Figure 1: Transistor as a switch

VIII Required Resources/apparatus/equipment with specification:

Sr. No.	Instrument/Components	Specification	Quantity
1	CRO	30MHz	1
2	Function Generator	3MHz	1
3	Dual regulated power Supply	0-30V	1
4	Digital Multimeter	DigitalMultimeter:3½ digit display.	1
5	Transister	BC107 or any other equivalent	1
6	Resister	1KΩ(0.5watts/0.25watts)	2
7	LED	3mm,Red/Green	1
8	Breadboard	5.5CM X 17CM	1
9	Connecting wires	Single strand Teflon coating(0.6mm diameter)	As per requirement

IX Precautions to be followed:

1. Do not switch ON power supply unless you have checked the circuit connections as per circuit diagram.
2. While doing Practical do not exceed the input voltage of the transistor beyond its rated voltage. This may lead to damaging of transistor.
3. Connection voltmeter and ammeter in correct polarities as shown in the circuit diagram.

X Procedure:

1. Connect the circuit as shown in Figure wave to the base.
2. Apply 5V, 1 kHz square wave to the base.
3. Apply +5V dc voltage to collector.
4. Observe the input and output wave forms simultaneously using oscilloscope.
5. Apply 0V square wave to the base.
6. Observe the input and output wave forms simultaneously using oscilloscope.

XI Required Resources/apparatus/equipment with specifications:

S. No	Instruments/Components	Suggested broad specification	Quantity
1			
2			
3			
4			

XII Actual procedure followed:

[illegible]

XIII Observations and Calculations:

Sr.No.	Region	Input Voltage	Output Voltage	Switch ON\OFF
1.	Cut-off	0V		
2.	Saturation	5V		

XIV Results:

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XV Interpretation of Results:

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XVI Conclusions &Recommendations:

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XVII Practical Related Questions:

1. Compare the theoretical and practical values of parameters in cut off and saturation mode.
2. List applications of transistor as a switch.
3. State the operating regions in which a transistor acts as a switch.

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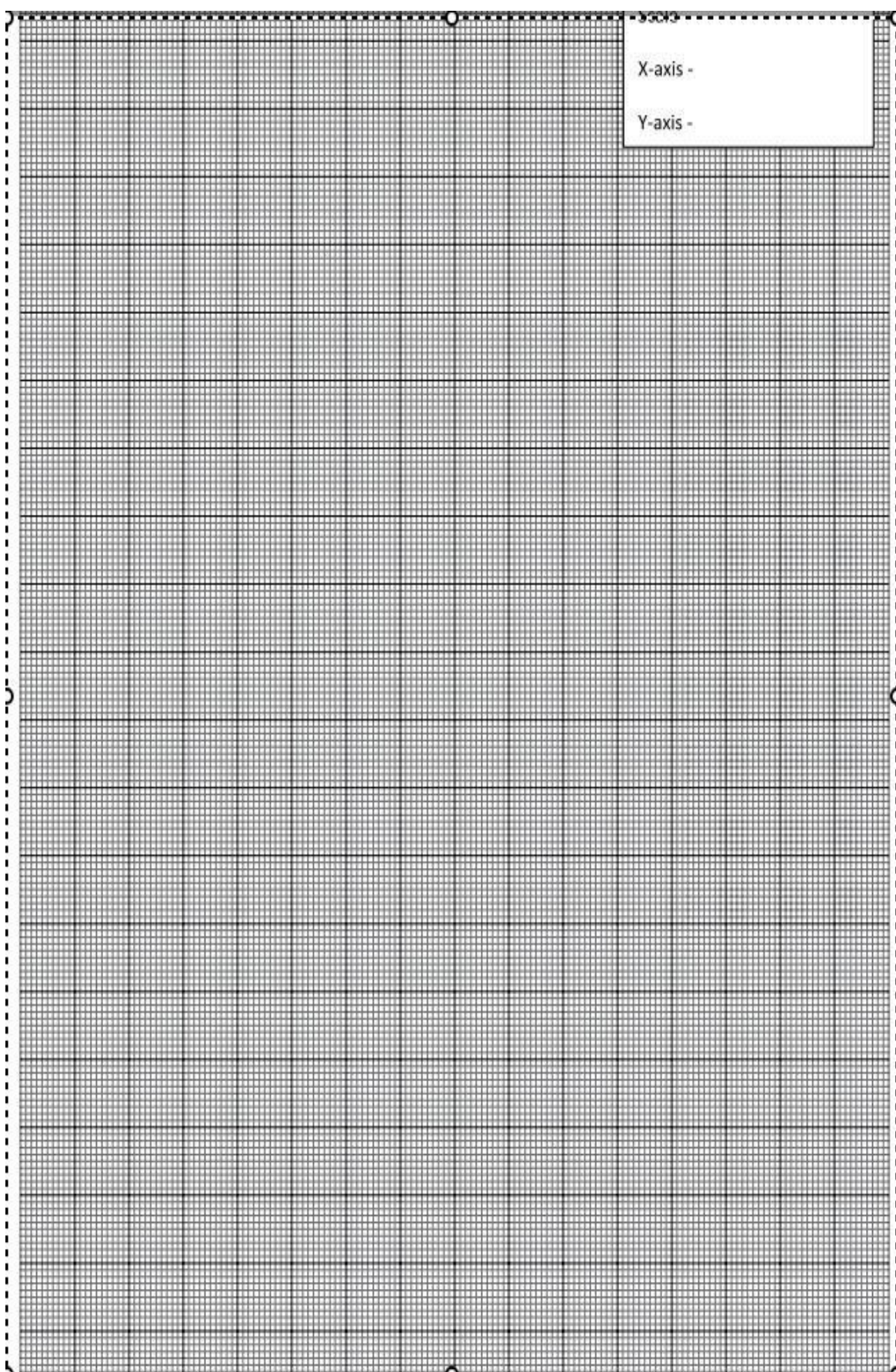
[illegible]

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XVIII References/ suggestions for further reading; includes websites:

1. <https://www.electrical4u.com/transistor-characteristics/>
2. <http://nptel.ac.in/courses/117107095/11>

Marks Obtained			Datedsignatureofteacher
Process Related (15)	Product Related(10)	Total (25)	



X-axis -

Y-axis -

PracticalNo.7 : Measure temperature of liquid using RTD (PT-100) transducer

I Practical Significance:

RTD (Resistance Temperature Detector) is most linear passive temperature transducer. PT-100 is most common low cost RTD. It is made up of platinum and it have 100 Ohm resistance at zero degree temperature. Since RTD have higher accuracy and repeatability, they are slowly replacing thermocouple in industrial application below 600° C. This practical will help you to measure temperature using RTD for given liquid.

II Industry/Employer Expected Outcome(s):

In the industry environment industrial electronics expected to handle various transducers for measurement of process parameters such as temperature, pressure, level, flow, displacement etc. It is necessary to measure temperature of different ranges in various industries. Employee should able to select, test and use appropriate temperature sensor.

III Course Level Learning Outcome(s):

Use various types of sensors and transducers.

IV Laboratory Learning Outcome(s):

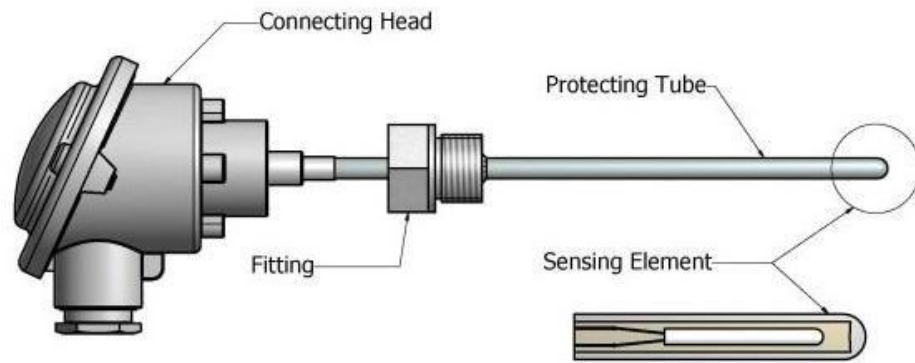
Use of RTD(PT-100) for measurement of temperature.

V Relevant Affective Domain related outcome(s):

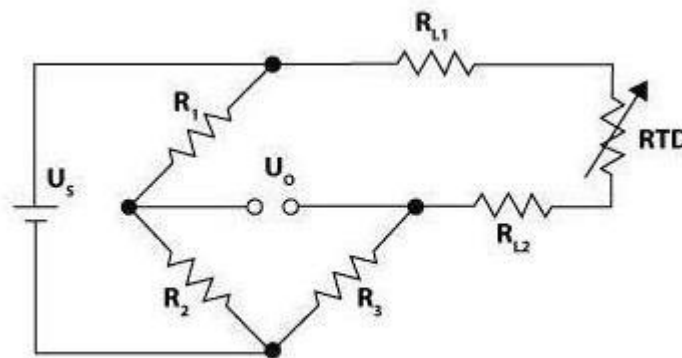
1. Follow safe practices.
2. Demonstrate working as a leader/a team member.
3. Maintain tools and equipment.

VI Relevant Theoretical Background(With diagrams if required):

Resistance Temperature Detector (RTD): The resistance of certain metals changes with temperature change. With the increase of temperature electrical resistance of certain metal increase in direct proportion to the rise of temperature. RTD is a sensor to measure temperature. Many RTD elements consist of a length of fine wire wrapped around a ceramic or glass core but other constructions are also used. RTD wire is made of pure materials of platinum, Nickel or copper. The material has accurate resistance/temperature relationship which is to provide an indication of temperature.



VII Actual Circuit diagram used in laboratory with equipment Specifications:



VIII Required Resources/apparatus/equipment with specification:

Sr.No.	Name of Resource	Suggested Board Specifications	Quantity
1	RTD setup	Any setup with standard RTD	1No.
2	DC power supply	0-30V,0-2A	1No.
3	MULTIMETER	Digital	1No.

IX Precautions to be followed:

1. Ensure proper earthing to the equipment.
2. Ensure that the Power Supply switch is in off condition initially.
3. Ensure proper settings of range of analog meter before use

X Procedure:

1. Connect RTD in bridge circuit as shown in setup.
2. Place mercury thermometer in given liquid.
3. Place RTD's metal part in the same liquid.
4. Measure resistance of RTD and output voltage for each 10° rise in temperature keeping heater on.
5. Repeat the procedure for 10 times.
6. Plot the Graph.

XI Required Resources/apparatus/equipment with specifications:

S. No	Instruments/Components	Suggested broad specification	Quantity
1			
2			
3			
4			

XII Actual procedure followed:

[illegible]

XIII Observations and Calculations:

SR.NO.	Temperature in Degree	Output Voltage	Resistance in ohm

XIV Results:

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XV Interpretation of Results:

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XVI Conclusions & Recommendations:

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XVII Practical Related Questions:

1. List standard type of RTD available in market.
2. State method of measurement of resistance of RTD.
3. State Practical reason to cover RTD with metal case.
4. State the Meaning of PT-100.
5. State resistance of RTD at Room temperature

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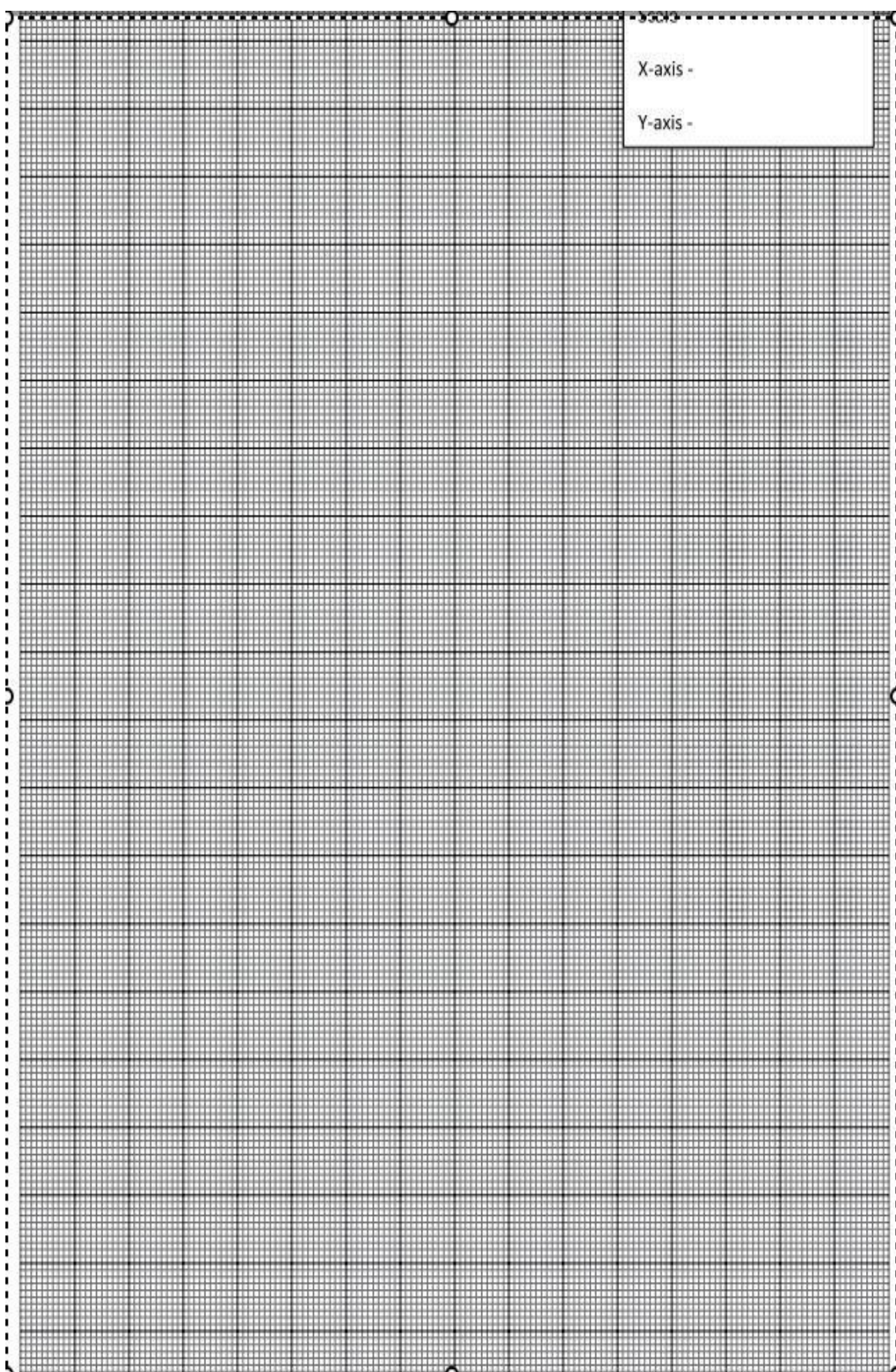
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XVIII References/suggestions for further reading;includes websites:

1. Laboratory Manual for Introductory Electronics Practicals, Maheshwari, L.K.; Anand, M.M.S., New Age International Pvt. Ltd. New Delhi: ISBN: 9780852265543
2. APPLICATION NOTES-PRACTICAL TEMPERATURE MEASUREMENTS from OMEGA
3. <https://youtube/WmIgusHZyPc?t=42>

Marks Obtained			Dated signature of teacher
Process Related (15)	Product Related(10)	Total (25)	



X-axis -

Y-axis -

PracticalNo.8: Measure the resistance of LDR in varying light intensity. I

Practical Significance:

LDR (Light Dependent Resistor) sensor modules are used where there is a need to sense the presence and absence of light is necessary. These resistors are used as light sensors and the applications of LDR mainly include alarm clocks, automatic street light control, light intensity meters and burglar alarm circuits.

II Industry/Employer Expected Outcome(s):

Many industries which manufacture security systems. LUX meter, object counting instruments uses LDR as photo sensitive transducer. Employee should be able to select, test and use proper photo sensing devices as per the application.

III Course Level Learning Outcome(s):

Use various types of sensors and transducers.

IV Laboratory Learning Outcome(s):

Use passive transducer to measure resistance.

V Relevant Affective Domain related outcome(s):

1. Follow safe practices.
2. Demonstrate work in as a leader/a team member.
3. Maintain tools and equipment.

VI Relevant Theoretical Background (With diagrams if required):

Light dependent resistors or in other words photo resistors are very useful especially in light/dark sensor circuits. They can also be referred to as photoconductors. Normally the resistance of an LDR is very high, but when they are illuminated with light resistance drops dramatically. A LDRs can have a variety of resistances and functions. For example, it can be used to turn on a light when the LDR is in darkness or to turn off a light when the LDR is in light. It can also work the other way around so when the LDR is in light it turns on the circuit and when it's in darkness the resistance increase and disrupts the circuit.

VII Actual Circuit diagram used in laboratory with equipment Specifications:

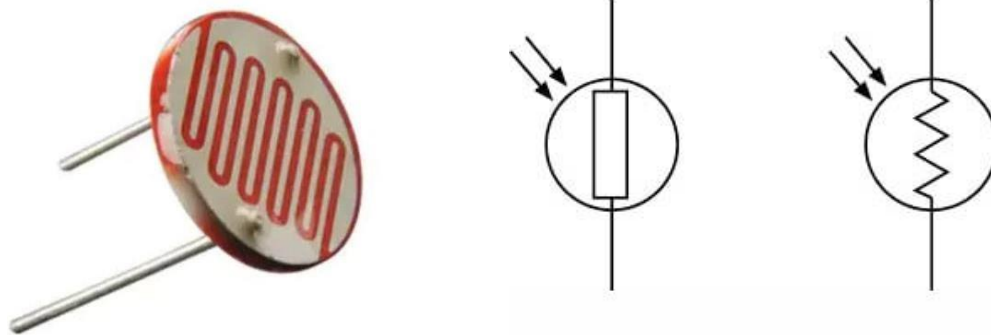


Figure1:Symbol

I. Practical setup

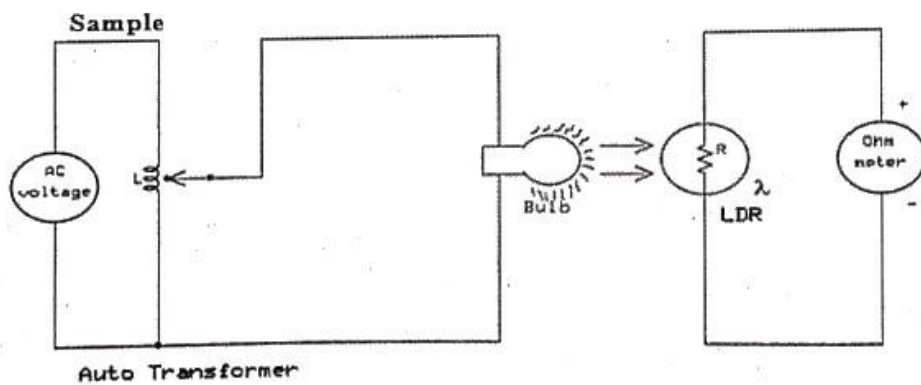


Figure 2 :- To measure the Resistance variation with light

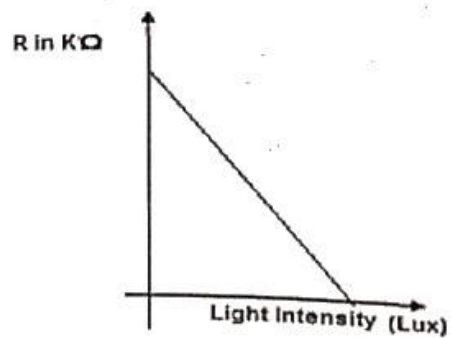


Figure 3.Ideal Characteristics of LDR

VIII Required Resources/apparatus/equipment with specification:

S.No.	Instrument/Components	Specification	Quantity
1	Digital Multi meter	DigitalMultimeter:3½digitdisplay	1
2	Bulb	60wor100woranyotherwatt	1
3	LDR		1
4	Connecting wires	Single stand Teflon coating(0.6mm diameter)	As per requirement

IX Precautions to be followed:

1. Do not switch ON power supply unless you have checked the circuit connections as per the circuit diagram.
2. Connect voltmeter and ammeter in correct polarities as shown in the circuit diagram.

X Procedure:

1. Connect the circuit as shown in figure2
2. Connect the ohm-meter section of DMM to LDR as shown in figure2.
3. Keep the LDR in front of bulb in such way that maximum light falls on LDR.
4. Initially adjust the output voltage of dimmer stat to zero volt.
5. Measure the resistance of LDR.
6. Now increase the dimmer stat voltage in steps of 30 V.
7. Repeat steps 5 and 6 till the output voltage of dimmer statreaches upto230volt.
8. Plot the graph light intensity (bulb voltage) verses resistance of LDR.
9. Take light intensity (bulb voltage)on X-axis and LDR resistance on Y-axis.
10. Plot the graph.

XI Required Resources/apparatus/equipment with specifications:

S. No	Instruments/Components	Suggested broad specification	Quantity
1			
2			
3			
4			

XII Actual procedure followed:

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XIII Observations and Calculations:

Measure resistance(R) in dark condition=.....k Ω

Sr. No.	Bulb voltage/light intensity(V/Lux)	ResistanceR(Ω)
1		
2		
3		
4		
5		
6		

XIV Results:

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XV Interpretation of Results:

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XVI Conclusions &Recommendations:

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XVII Practical Related Questions:

1. Which materials are used to fabricate LDR?
2. List four applications of LDR.
3. Whether the LDR is passive or active component?
4. Draw the symbol of LDR.
5. What is dark current related to LDR?
6. At what intensity of light, resistance of LDR is minimum? What is its value?

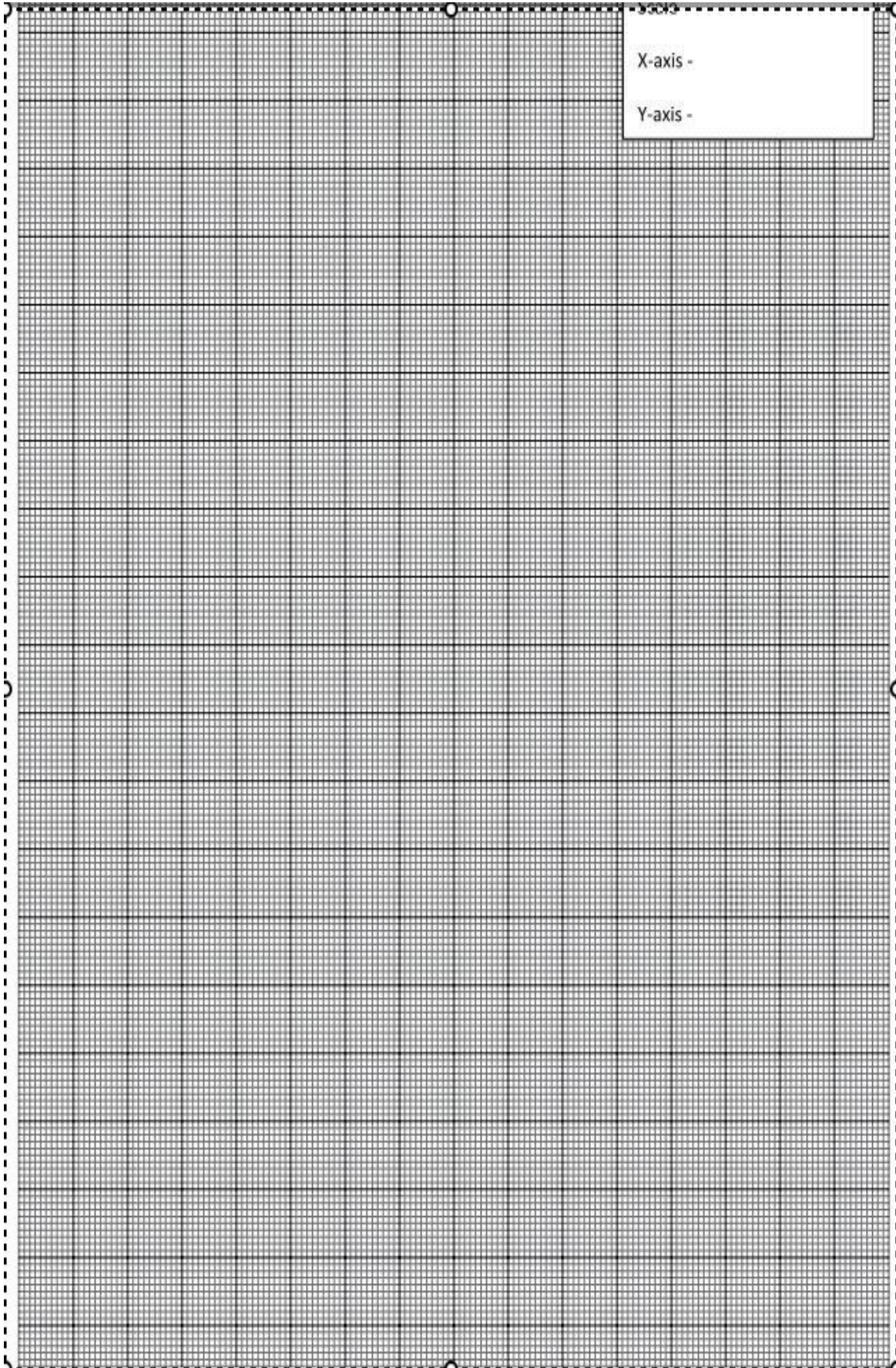
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XVIII References/suggestions for further reading; includes websites:

1. Laboratory Manual for Introductory Electronics Practicals, Maheshwari, L.K.; Anand, M.M.S., New Age International Pvt. Ltd. New Delhi: ISBN: 9780852265543
2. APPLICATIONNOTES-PRACTICAL TEMPERATURE MEASUREMENTS from OMEGA
3. <https://youtube/WmIgusHZyPc?t=42>

Marks Obtained			Dated signature of teacher
Process Related (15)	Product Related(10)	Total (25)	



X-axis -

Y-axis -

